

INDIVIDUAL COURSEWORK

Introduction to Linux · CS Year 2

Shell Scripting Assignment

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 TASK 19 OF 22

Dead Process Restarter

 **Script** deadproc.sh

 **Time** 40 min

 **Difficulty**
Advanced

 **Marks** 100

Skills Covered in This Task

 ps aux · grep · \$? <i>Check if running</i>	 functions <i>restart_proc()</i>	 if / else <i>Branch on status</i>	 >> · date <i>Audit log</i>
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Scenario

A critical background service sometimes crashes. Write a script that checks if a given service (by process name) is running, attempts to restart it using a given command if it is not, and logs all events.

Requirements

Your script `deadproc.sh` must satisfy all of the following:

#	Requirement	Skills	Marks
1	Accept two arguments: process name (\$1) and restart command (\$2). Exit with usage if wrong.	<code>\$# · exit</code>	10
2	Define a <code>log()</code> function that writes "[timestamp] level: message" to <code>/tmp/restarter.log</code> .	<code>function · >> · date</code>	15
3	Check if \$1 is currently running using <code>ps aux grep -v grep grep</code> .	<code>ps aux grep -v grep grep -q</code>	20
4	If running: log INFO and print "OK: <process> is running".	<code>log() · echo</code>	10
5	If NOT running: log WARN, print the warning, run \$2 in background (&), and log the restart.	<code>\$2 & · log() · &</code>	25
6	After attempting restart: sleep 2 and re-check. Log SUCCESS or FAILED.	<code>sleep · \$?</code>	20

Expected Output

```
$ ./deadproc.sh nginx 'nginx -g daemon off;'

[WARN] nginx is NOT running

Attempting restart...

Restart command launched

[SUCCESS] nginx is now running

Log saved to /tmp/restarter.log
```

Hints

Check	<code>ps aux grep -v grep grep -q "\$1" ; STATUS=\$?</code>
Background	<code>\$2 > /dev/null 2>&1 &</code>
Re-check	<code>sleep 2 ; ps aux grep -v grep grep -q "\$1" && log SUCCESS ... log FAILED ...</code>
Log fn	<code>log() { echo "[\$(date '+%Y-%m-%d %H:%M:%S')] \$1: \$2" >> /tmp/ restarter.log; }</code>

Submission Checklist

☐	Script starts with <code>#!/bin/bash</code>
☐	Script is executable: <code>chmod +x deadproc.sh</code>
☐	Script runs without errors on Webminal
☐	All requirements are met (tick each one above)
☐	Variables are quoted: <code>"\$var"</code> throughout
☐	Script includes comments (<code>#</code>) explaining key sections
☐	Tested with valid input and tested with invalid/missing input

Grading Rubric

Criteria	Full marks	Awarded
Argument validation and log function	25	/ 25
Process detection with <code>ps + grep -v grep</code>	25	/ 25
Background restart and re-check	30	/ 30
Complete audit log	20	/ 20
TOTAL	100	/ 100